

Annex B

Cover Page for CJC 2009 Maths H2 Prelim Exam Papers

Qn/No	Topic Set	Paper Number	Answers	Marks
1	Area / Volume	9740/01	$\frac{25 + 12\sqrt{2}}{2}$	4
2	Mathematical Induction	9740/01	(i) 4, 18, 48, 100 (ii) $\sum_{r=1}^n u_r = n(n+1)^2$	7
3	Vectors - Basics	9740/01	(i) $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ (ii) $\overrightarrow{OM} \cdot \overrightarrow{AB} = 0$ (iii) $6\sqrt{14}$ units ²	6
4	Differential Equations	9740/01	$y = \frac{k(x+2)^3 + 1}{3x}$	7
5	Techniques of Integration	9740/01	(i) $\sin x - x \cos x + c$ (ii) $-2 \ln \left(\frac{1}{\sqrt{x}} + \sqrt{\frac{1-x}{x}} \right) + c$	6
6	Maclaurin's Series	9740/01	$e + ex^2 + \frac{e}{2}x^4 + \dots$ 1.388 (3dp)	10
7	Maxima / Minima Problems	9740/01	\$4.92	10
8	Complex Numbers - Algebraic	9740/01	Statement is true if p is real. (a)(i) $p = 5$ (a)(ii) $-\frac{1}{2} \pm \frac{\sqrt{3}}{2}i, 1 \pm 2i$ (b) $\frac{3\pi}{4}$ radian	11
9	Differentiation – • Curve Sketching • Transformation	9740/01	(i) $(4a, \frac{\sqrt{a}}{a}), (4a, -\frac{\sqrt{a}}{a})$ (iv) Method 1: A translation of $4a$ unit in the direction of positive x -axis followed by a scaling of $1/3$ units along the x -axis. Method 2: Scaling of $1/3$ units along the x -axis followed by a translation of $4a/3$ unit in the direction of positive x -axis.	11
10	Arithmetic / Geometric Progression	9740/01	(a)(i) 8 (ii) \$41 (b)(i) $d = b \ln 2$ (ii) $= \frac{n \ln 2}{2} [(n+1)b + 4]$ (iii) $-0.194 \leq b \leq -0.0913$	14
11	Vectors – Lines & Planes	9740/01	(ii) 75.7°	14

			(iii) $\sqrt{22}$ (iv) $b = 2a + 4$, where $a \neq 2$ (v) $b = 6$ or 10	
	Pure Maths			
1	Partial Fraction / Sigma Notation	9740/02	(i) $\frac{2}{r} + \frac{-3}{r-1} + \frac{1}{r-2}$ (ii) $\frac{n-2}{n(n-1)}$ (iii) $\frac{n}{(n+2)(n+1)}$	9
2	Area / Volume	9740/02	(a) 1.68 (b) $b = \frac{5}{2}$	9
3	Complex Numbers	9740/02	(a) $z = \sqrt{2}e^{i\frac{\pi}{8}(2k+\frac{5}{6})}$, $k = 0, \pm 1, \pm 2, \pm 3, -4$ (b) Square	11
4	Functions	9740/02	(ii) $f^{-1} : x \rightarrow \frac{x}{2} + \frac{1}{2}\sqrt{x^2 + 4}$, $x \in (-\infty, \infty)$ (iv) $D_f : (0, \infty)$ and $R_g : [-1, 1]$. (v) Largest $D_g : (0, \pi)$ $fg : x \rightarrow \sin x - \frac{1}{\sin x}$, $x \in (0, \pi)$.	11
	Statistics			
5	Sampling Methods	9740/02	(i) From the school's registration list, randomly select a student from the first 5 on the list as a starting point. Select every 5 th student down the list from that starting point, until 300 names are obtained. (ii) A sampling frame is easily accessible. The population can be divided into non-overlapping subgroups	4
6	Hypothesis Testing	9740/02	p-value = 0.0771 Insufficient evidence at 5% level to indicate that the mean battery life of a iPhone is less than 120 hours. Probability of concluding the mean battery life of a iPhone is less than 120 hours when it is actually 120 hours is 0.05.	7
7	Probability	9740/02	(a) A and B are not mutually exclusive (b)(i) $\frac{11}{216}$ (ii) $\frac{3}{7}$	8
8	Correlation and Linear Regression	9740/02	(ii) Model (b): $y = ax^b$ Reason: The graph of $y = ax^b$ fits the scatter diagram better. (iii) $r = 0.932$ $a = 66.1$ $b = 0.306$ (iv) $x = 5.29$. Extrapolation, hence not reliable.	11

9	Binomial Distribution	9740/02	(i) 0.0124 (ii) 46 (iii) 0.801	9
10	Permutation and Combination	9740/02	(a)(i)210 (ii)207 (b)(i) 57600 (ii) 3628800	8
11	Normal Distribution	9740/02	(a) 0.916 (b)(i) 0.781 (b)(ii) No need to use Central Limit Theorem because the population follows a Normal Distribution. (c)(i) 0.159 (c)(ii) 0.998	13